

Jeremy R. Ash

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Experience

Excelsior Sciences

New York, NY

Oct 2025 – Pres

MAY 2026 – PRES AI/ML CONSULTANT
OCT 2025 – MAY 2026 PRINCIPAL SCIENTIST, MACHINE LEARNING DRUG DISCOVERY

- Leading ML platform development for Excelsior's automated DMTA (Design-Make-Test-Analyze) drug discovery projects, where novel automated chemical synthesis capabilities enable AI-guided exploration of ultra-large combinatorial libraries.
- Designing active learning workflows for multi-parameter optimization across potency and ADME objectives. Workflows are selecting compounds for automated chemical synthesis and biological assays on multiple proof of concept projects.
- Developing uncertainty quantification methods for deep learning-based chemical property prediction models; developing conformal prediction and domain adaptation methods to improve reliability under covariate shift.

Johnson & Johnson AI/ML Drug Discovery Data Science

Philadelphia, PA

Feb 2022 – Oct 2025

MAY 2025 – OCT 2025 PRINCIPAL SCIENTIST
JUL 2024 – MAY 2025 SENIOR SCIENTIST
FEB 2022 – JUL 2024 SCIENTIST

- Led development of an active learning platform, collaborating with a team of computational and experimental scientists automating the DMTA cycle for small molecules.
- Led development of an active learning workflow to optimize data production for ADMET models, resulting in >40% reduction in experimental data expenditure across multiple ADMET endpoints used by the entire small molecule portfolio. Developed Spotfire-based tool for tracking the performance of model selections.
- Led development of virtual screening workflows that discovered novel progressible series which advanced to hit-to-lead in multiple TA projects. Series were confirmed in SPR and crystallography. Developed workflows for the following models: DNA-Encoded Library (DEL)-based machine learning models, high content imaging-based deep learning models, and chemical structure-based graphormer models.
- Contributed to development of an end-to-end MLOps platform for ADMET prediction that deployed "global" models in LiveDesign, providing real-time support for all small molecule therapeutic area (TA) projects (50+ projects). Models use transformer-based molecular representations.
- Developed a predictive model evaluation tool in LiveDesign with a user-friendly React front end, supporting an initiative to improve model usage by TA projects. Project teams used the tool extensively to assess when and where to apply their models, substantially improving model usage.
- Managed contractors and IT to establish infrastructure for MLOps and data engineering.

SAS Institute JMP Stats R&D

Cary, NC

Aug 2018 – Feb 2022

OCT 2021 – FEB 2022 RESEARCH STATISTICIAN DEVELOPER
JAN 2020 – OCT 2021 RESEARCH STATISTICIAN TESTER
AUG 2018 – DEC 2019 GRADUATE INDUSTRIAL TRAINEE

- Researched, prototyped and evaluated statistical methods for JMP software using R, Python, SAS, and JSL (JMP scripting language).
- Supported development by writing and managing suites of unit tests covering JMP's statistical methods.
- Developed statistical methods and other features in JMP using C++.
- Selected projects: wavelet-based functional data analysis and design of experiments for chemometrics, multivariate control chart platform for high-dimensional process control, method for avoiding extrapolation in JMP's prediction profilers for machine learning models.

North Carolina State University Departments of Statistics and Bioinformatics

Raleigh, NC

Aug 2014 – Dec 2019

GRADUATE RESEARCH ASSISTANT | ADVISORS: JACQUELINE HUGHES-OLIVER AND DENIS FOURCHES

- Developed statistical methods for evaluation of virtual screening models in drug discovery.
- Developed descriptors based on molecular dynamics simulations to improve QSAR machine learning models.
- Developed chemmodlab, an R package for building cheminformatics machine learning models.

Louisiana State University Department of Biology

Baton Rouge, LA

Feb 2014 – Aug 2014

RESEARCH ASSOCIATE | ADVISOR: JEREMY BROWN

- Collaborated with the lab of Kyle Gallivan (Florida State University) developing TreeScaper, software for analysis of phylogenetic networks, providing novel community detection methods and nonlinear dimensionality reduction.

Skills

Core Competencies	AI/ML for drug discovery, active learning, statistical modeling, ADMET modeling, virtual screening, cross-functional collaboration
ML & Data Science	Python ML toolkit (PyTorch, scikit-learn, chemprop), large data engineering (polars, pandas), deep learning and transformer architectures, uncertainty quantification, data visualization
Comp Chem	Cheminformatics methods, RDKit, Schrödinger Suite, KNIME, Spotfire, QSAR, DNA-encoded libraries, molecular dynamics
Programming	Python, R, C++, SQL, AWS (SageMaker, S3, Athena, EC2), Prefect, Pulumi, Docker, CI/CD (Gitlab), Claude Code

Education

North Carolina State University

BIOINFORMATICS, PH.D. | STATISTICS, M.S.

Raleigh, NC

Aug 2014 - Dec 2019

The University of Texas at Austin

BIOLOGY, B.S.

Austin, TX

Aug 2010 - Aug 2013

Selected Publications

Google Scholar: <https://scholar.google.com/citations?user=Au7CUSAAAAAJ&hl=en&oi=ao>

ORCID: <https://orcid.org/0000-0002-8041-8524>

- 1) **Ash JR**, Wognum C, Rodríguez-Pérez R, Aldeghi M, Cheng AC, Clevert D-A, Engkvist O, Fang C, Price DJ, Hughes-Oliver JM, Walters WP. *Practically significant method comparison protocols for machine learning in small molecule drug discovery*. Journal of Chemical Information and Modeling, 2025, 65 (18), 9398-9411.
- 2) Van Rompaey D, Ray Chaudhuri S, Ahmad M, Cisar J, Van Den Bergh A, **Ash JR**, Wu Z, Bryan MC, Edwards JP, DesJarlais R, Wegner JK. *Toward Dose Prediction at Point of Design*. Journal of Medicinal Chemistry, 2024, 67 (24), 22282-22290.
- 3) Wognum C, **Ash JR**, Aldeghi M, Rodríguez-Pérez R, Fang C, Cheng AC, Price DJ, Clevert D-A, Engkvist O, Walters WP. *A call for an industry-led initiative to critically assess machine learning for real-world drug discovery*. Nature Machine Intelligence, 2024, 6 (10), 1120-1121.
- 4) **Ash JR**, Hughes-Oliver JM. *Confidence bands and hypothesis tests for hit enrichment curves*. Journal of Cheminformatics, 2022, 14(1), 50.
- 5) **Ash JR**, King C, Lancaster L, Lekivetz R, Morgan J, Saanchi Y. *Testing the prediction profiler with disallowed combinations—A statistical engineering case study*. Quality Engineering, 2022, 34(4), 507-521.
- 6) Odenkirk MT, Zin PPK, **Ash JR**, Reif DM, Fourches D, Baker ES. *Structural-based connectivity and omic phenotype evaluations (SCOPE): a cheminformatics toolbox for investigating lipidomic changes in complex systems*. Analyst, 2020, 145 (22), 7197-7209. (Cover Article)
- 7) **Ash JR**, Kuenemann MA, Rotroff D, Motsinger-Reif A, Fourches D. *Cheminformatics Approach to Exploring and Modeling Trait-Associated Metabolite Profiles*. Journal of Cheminformatics, 2019, 11 (1), 43.
- 8) **Ash JR**, Hughes-Oliver JM. *Chemmodlab: A Cheminformatics Modeling Laboratory R Package for Fitting and Assessing Machine Learning Models*. Journal of Cheminformatics, 2018, 10 (1).
- 9) **Ash JR**, Fourches D. *Characterizing the Chemical Space of ERK2 Kinase Inhibitors Using Descriptors Computed from Molecular Dynamics Trajectories*. Journal of Chemical Information and Modeling, 2017, 57 (6), 1286-1299.
- 10) Huang W, Zhou G, Marchand M, **Ash JR**, Morris D, Van Dooren P, Brown JM, Gallivan KA, Wilgenbusch JC. *TreeScaper: Visualizing and Extracting Phylogenetic Signal from Sets of Trees*. Molecular Biology and Evolution, 2016, 33 (12), 3314-3316.

Patents

- 1) **Ash JR**, Ceulemans H, Allen S. *Systems and methods for drug discovery using cellular imaging and predictive models*. Non-provisional patent application filed Jan 31, 2025.
- 2) Parker RJ, Barker CA, **Ash JR**, Gotwalt CM. *Analytic system for interactive graphical model selection based on wavelet coefficients*. US Patent 11,361,255 B2. 2021.
- 3) **Ash JR**, Gotwalt CM, Lancaster LC. *Analytic system with extrapolation control in interactive graphical prediction evaluation*. US Patent 10,963,804. 2021.

Awards

- 1) ACS Editor's Choice for excellence in publication, *Practically Significant Method Comparison Protocols for Machine Learning in Small Molecule Drug Discovery*, 2025.
- 2) ACS Division of Chemical Information Scholarship of Excellence for poster presentation, *Molecular Modeling of Differential ERK1/2-Ligand Dynamic Interactions*, 2018.
- 3) ACS Division of Chemical Information Scholarship of Excellence for poster presentation, *Cheminformatics Approach to Exploring and Modeling Trait-Associated Metabolite Profiles*, 2017.
- 4) ACS Editor's Choice for excellence in publication, *Characterizing the Chemical Space of ERK2 Kinase Inhibitors Using Descriptors Computed from Molecular Dynamics Trajectories*, 2017.
- 5) Triangle Center of Evolutionary Medicine Fellowship, 2017.
- 6) NCSU Graduate School Fellowship, 2014.

Selected Talks

- 1) *Practically significant method comparison protocols for machine learning in small-molecule drug discovery*. Polaris Webinar, Virtual 2024 [\[Video\]](#).
- 2) *Evaluating performance of global ADMET models for estimating properties within a drug discovery project's chemical series*. ACS Conference, Denver, CO 2024 [\[DOI\]](#).
- 3) *Data to decisions: Machine learning and data visualization tools optimize compound selection for solubility predictive models in real-time support of small molecule discovery*. ACS Conference, New Orleans, LA 2023 [\[DOI\]](#).
- 4) *Analyzing Spectral Data: Preprocessing and Beyond*. JMP Discovery Summit, Virtual 2021.
- 5) *Methods for Helping Users Avoid Extrapolation when Making Predictions with Statistical and Machine Learning Models*. Joint Statistical Meetings and JMP Discovery Summit, Virtual 2020.
- 6) *Fault Detection and Diagnosis of the Tennessee Eastman Process using Multivariate Control Charts*. JMP Discovery Summit, Virtual 2020.
- 7) *Structure-Based Approach to Exploring and Modeling Trait-Associated Metabolite Profiles*. ACS Conference, Boston, MA 2018 [\[DOI\]](#).
- 8) *Characterizing the Chemical Space of Kinase Inhibitors Using Molecular Descriptors Computed from MD Trajectories*. ACS Conference, Washington, DC 2017 [\[DOI\]](#).

Service

- 2018-Pres **Reviewer**, Scientific journals (e.g. JCIM, Journal of Cheminformatics)
- 2018 **Organizer**, First Annual NCSU Genomic Sciences and Biomath Research Symposium
- 2017-2018 **President**, NCSU Genomic Sciences Graduate Student Association

References

Available upon request